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BILLET-T3 Multi-Channel Spectral Processing Engine

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:13 UTC+8 Beijing Time)

Abstract

BILLET-T3 is a synthesizable SystemVerilog spectral-processing IP block for four independently configured complex-sample channels. A shared serial engine supports 16-, 32-, 64-, 128- and 256-point FFT/IFFT, programmable stage scaling and block floating point, with rectangular, Hann or Hamming windowing and non-overlapping or 50%-overlapping frame assembly. Independent verification passed 217 complete transform frames comprising 23,040 bins, plus 44 standalone channel-buffer frames comprising 2,624 sample transfers. Additional checks covered reset-aborted output, arithmetic units, legacy compatibility and checker fault sensitivity. Generic synthesis confirmed the required reachable hierarchy and absence of inferred latches. These results establish bounded RTL functionality and synthesizability, not physical implementation performance. [E021, E022, E023]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

Four raw-sample rings decouple interleaved channel acquisition from one radix-2 decimation-in-time transform engine. Keeping raw history permits each overlapping frame to apply its own chronological window coefficients. One reservation per channel prevents overwrite until transfer into the core completes. This conservative ownership scheme avoids copying retained samples and permits other eligible channels to accept input while one channel is blocked. Shared arithmetic reduces replication but limits aggregate throughput; four contexts do not imply four simultaneous transforms. Primary sample storage totals 6,272 logical bytes: 4,096 bytes of rings, 1,024 bytes of normalized transform storage and 1,152 bytes of raw butterfly storage, excluding coefficients and control. These capacities are architectural, not technology area measurements. [E001, E006, E007]

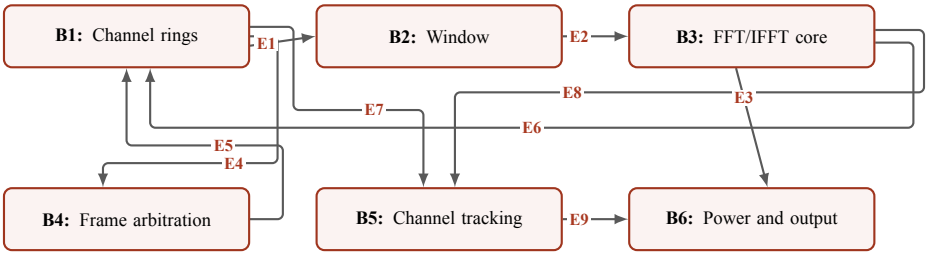


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Channel rings	Instance channel_buf captures per-channel configuration, stores four 256-word raw complex rings, counts accepted samples and protects complete-frame reservations.
B2: Window	Instance window applies combinational rectangular bypass or symmetric Q1.15 Hann/Hamming multiplication at the chronological frame index.

Block	Responsibility
B3: FFT/IFFT core	Instance fft_core performs bit-reversed loading, serial butterflies, whole-stage normalization and natural-order output. Literal children are bitrev, twiddle_rom, butterfly and bfp_scaler.
B4: Frame arbitration	Logic inside channel_buf selects pending frames round-robin, retains selection through stalls and advances priority after the final frame-transfer handshake.
B5: Channel tracking	Top-level registers capture the selected channel on its first core input transfer and retain separate active-output and completion tags.
B6: Power and output	Instance power_spectrum computes exact component-square sums. Complex bins, power, exponent and channel tag share the core's output handshake without an additional pipeline stage.

Directed relationship	Meaning
E1: B1 to B2	Chronological raw samples, index and captured configuration
E2: B2 to B3	Windowed complex samples; no registered latency
E3: B3 to B6	Post-normalization complex bins and block exponent
E4: B1 to B4	Pending reservations and frame metadata
E5: B4 to B1	Selected read context and reservation release
E6: B3 to B1	Core input readiness controls frame transfer
E7: B1 to B5	Channel tag on first accepted frame beat
E8: B3 to B5	Final-output handshake captures completion identity
E9: B5 to B6	Output and completion channel tags

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Reset forces every public output to zero without requiring a rising edge.
in_valid, in_last, in_re, in_im, ch_id	Input	1, 1, signed 16, signed 16, 2 bits	Offered complex sample and channel 0 through 3. Only in_valid && in_ready consumes a sample. in_ last is informational and does not delimit frames.
cfg_log2_n, cfg_ifft	Input	4, 1 bits	Transform length exponent 4 through 8; direction 0 selects FFT and 1 selects IFFT.
cfg_scale_mode, cfg_ stage_shift_mask	Input	2, 8 bits	Scaling mode 0 shifts every stage, 1 uses mask bit s for stage s, and 2 selects BFP. Mode 3 is invalid.
cfg_overlap_mode, cfg_ window_mode	Input	1, 2 bits	Overlap 0 selects hop N; 1 selects hop N/2. Window 0 is rectangular, 1 Hann and 2 Hamming; window 3 is invalid.
in_ready, cfg_error	Output	1 bit each	Readiness applies to the offered channel. Invalid configuration when capture is required asserts nonsticky cfg_error independently of in_valid and prevents acceptance.
out_ready	Input	1 bit	Sink acceptance control. A bin transfers only when out_valid && out_ready.

Signal / group	Direction	Width	Contract
out_valid, out_last, out_re, out_im, out_power	Output	1, 1, signed 16, signed 16, unsigned 32 bits	Valid complex bin and exact power; out_last marks bin N-1. The complete payload remains stable during a stall.
out_bin_idx, out_block_exp, out_ch_id	Output	8, 4, 2 bits	Natural-order bin index, frame-wide stage-shift count and channel identity. Exponent ranges from zero through log2(N).
frame_done, frame_done_ch_id	Output	1, 2 bits	One-cycle completion pulse immediately after the final output handshake, with completed channel identity. Completion identity is zero otherwise.
overflow_flag, saturation_count	Output	1, unsigned 32 bits	Sticky component-saturation flag and cumulative saturating count across frames and channels. Both clear on reset.

Parameter	Default	Meaning / supported values
Structural configuration	Fixed; no public structural parameters	Exactly four contexts, maximum N=256 and one serial transform core. There are no DEPTH or MAX_WORDS parameters.
Per-channel runtime configuration	No default active epoch	Every channel captures all configuration fields on its first accepted sample. Supported lengths are exclusively N=16, 32,64,128,256; legal direction, scaling, overlap and window encodings are defined above.
Epoch lifetime	Inactive after reset	Non-overlap epochs end on the Nth accepted sample and can be reconfigured after reservation release. Overlap configuration and retained history persist until reset.

III Behavior and corner cases

A Frame assembly and ownership

Non-overlap frame m contains samples m*N through m*N+N-1. Overlap frame m contains samples m*(N/2) through m*(N/2)+N-1. No frame exists before N accepted samples; subsequent overlap frames require exactly N/2 new samples. Rings wrap at the captured N, not their physical capacity. A pending or transferring frame blocks writes only to its own channel. Final transfer releases the reservation, retaining the newest N/2 raw samples for overlap or clearing fill for non-overlap. Same-edge reservation replacement is not supported. [E001, E006]

B Configuration and scheduling

Live pins describe only the offered channel. Once captured, subsequent changes, including invalid encodings, cannot alter or block that epoch. A non-overlap channel awaiting release can report invalid next-epoch configuration while its previous frame remains pending. Registered round-robin selection starts at channel 0 and advances to the channel following each completed frame transfer. Frames never interleave internally or at output; selection order, rather than global arrival order, determines output order. Eventual sink readiness is required for starvation-free progress. [E002, E006]

C Window arithmetic

For chronological index n, Hann uses w[n]=0.5-0.5*cos(2*pi*n/(N-1)); Hamming uses w[n]=0.54-0.46*cos(2*pi*n/(N-1)). Embedded coefficients are q[n]=min(32767,floor(32768*w[n]+0.5)), with exact mirrored symmetry. Each signed component is multiplied at full 32-bit precision and arithmetic-shifted right 15, truncating toward negative infinity. Rectangular mode bypasses multiplication exactly. Windowing precedes bit reversal and is reapplied to reused raw samples. [E011]

D Transform and normalization

The radix-2 DIT core preserves Gen-2 twiddle values, including positive unity 32767 and negative unity -32768. Signed 17-bit twiddle outputs preserve +32768 when IFFT negates the forward imaginary coefficient. Complex products use signed 33-bit products and 34-bit sums/differences before shifting right 15; raw butterfly results are signed 18-bit. After observing a complete stage, normalization applies its single selected arithmetic shift and clamps each component to [-32768,32767]. BFP shifts once if any raw component exceeds that range; extreme results can still saturate. IFFT adds no implicit normalization. [E003, E005, E007, E010]

E Power and numerical status

Power is exactly Re(X)^2+Im(X)^2 for the delivered, scaled and saturated components, with maximum 0x80000000 at (-32768,-32768). No exponent compensation is applied; unsaturated reconstructed-scale power additionally carries 2^(2*out_block_exp). Each clamped real or imaginary component increments saturation_ count, which saturates at 0xffffffff; windowing and power do not contribute. [E007, E008]

F Pre-edge and post-edge timing

Handshakes use pre-rising-edge values. A sample completing a frame at edge E creates its reservation only after E; registered selection first exposes it after E+1 or later. At final core-input handshake F, the core enters BUTTERFLY_READ. Two cycles per butterfly and two per normalization sample produce first out_valid immediately after F+3*N*log2(N)+1: 193 cycles for N=16 and 6,145 for N=256. After a nonfinal output transfer at O, the next bin becomes valid after O+1 and first transfers at O+2. Final transfer at L asserts frame_done after L; it clears after L+1. No new core-input beat transfers on L. Arbitration can add interframe gaps. [E002, E007]

G Reset and inactive outputs

Asynchronous reset aborts partial, pending, computing and stalled-output frames, clears history validity, configuration epochs, arbitration, tags and cumulative status, and prevents pre-reset results from reappearing. Sample memories need not reset because validity and counts prevent uninitialized reads. Outside reset, inactive output data, power, index, exponent, channel and last are zero; cumulative status remains visible. [E006, E007, E009]

IV Verification methods and measured results

A Exact transform arithmetic across supported configurations

Method. Independent integer oracle with mathematical coefficient generation and a recursive transform organization; numerical-matrix seed 1701 exercised five lengths, two directions, three scaling modes and three windows.

Result. All 90 frames and 8,928 bins passed exact comparison. The run included 18 saturating frames and BFP cases both with and without shifts. Detailed frame denominators and outcomes are retained in the behavior results ledger.

Evidence: E013, E015, E016, E021

B Independent numerical plausibility and legacy continuity

Method. Thirty low-amplitude integer-oracle versus floating-point DFT cases using seed 109; direct Gen-2/T3-core trace comparison using seed 8917.

Result. All DFT cases passed the log2(N)+2 LSB component-error limit; the largest observed error was 4.87109375 LSB. Legacy comparison passed 50 frames, 4,960 bins and 121,643 public-interface cycles with byte-identical traces.

Evidence: E013, E016, E021

C Overlap chronology, channel isolation and fair reservation handling

Method. Accepted-sample history scoreboards; overlap-repeated-wrap seed 291 and standalone channel-buffer-contract seed 51.

Result. Overlap passed 64 transform frames and 8,704 bins, including 32 repeated-wrap comparison events. Buffer verification passed 44 frames and 2,624 transfers, with 1,805 stalled frame cycles, 44 round-robin selections and 15 simultaneous input/frame-transfer events.

Evidence: E006, E015, E016, E021

D Reset, invalid configuration, stalls and completion timing

Method. Cycle-level boundary scenario, seed 73, with active X/Z rejection and pre/post-edge comparisons.

Result. Passed 13 complete frames and 448 bins, plus 15 accepted bins from a reset-aborted frame. Recorded 52 invalid idle and 52 invalid valid observations, five asynchronous resets without a clock, and first/middle/final-bin stalls. Across transform scenarios, 217 completion pulses followed final handshakes.

Evidence: E012, E016, E021

E Arithmetic-unit boundaries

Method. Independent directed and seeded unit vectors against delivered combinational units and scaler sequencing.

Result. Passed 5,952 window, 496 bit-reversal, 256 twiddle, 4,177 power and 4,096 butterfly vectors, plus scaler reset, observation, normalization and commit checks.

Evidence: E013, E016, E021

F Reachable synthesizable hardware and exact public interface

Method. Yosys 0.33 generic technology-independent synthesis without ABC, hierarchy and multiplier inspection, memory inventory, port checks and no-latch assertions.

Result. Passed all 28 public ports and required literal instance relationships. Pre-mapping sample memories contained 32,768 ring bits, 8,192 normalized-core bits and 9,216 raw-core bits. Eight multipliers were reachable; no inferred latches were detected.

Evidence: E016, E022

G Source-based delivery and checker sensitivity

Method. Compile retained source copies for a channel-3, N=16 impulse test; separately inject stuck-zero power into a disposable private copy, then rerun unchanged sources.

Result. Source-only reproduction passed one frame and 16 bins over 264 cycles. The unchanged rerun produced an identical trace. The private defect was rejected at bin 0 because zero power differed from expected 1,310, 720, despite simulator exit success. Production sources were not modified.

Evidence: E014, E016, E023

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	7 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 5	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

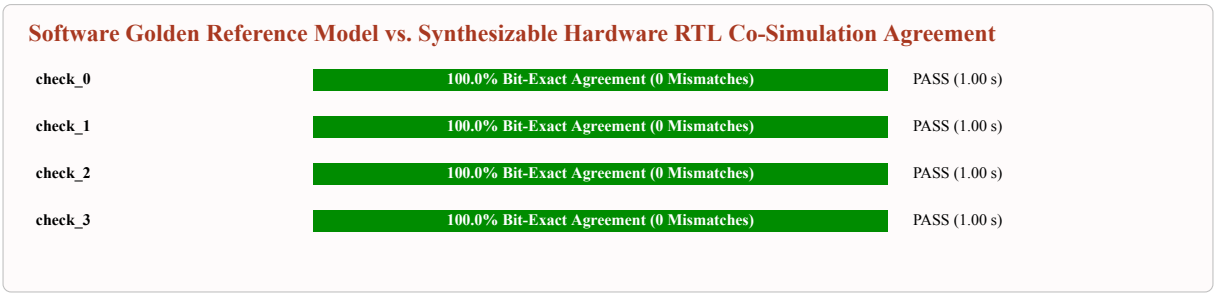


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +2.929 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

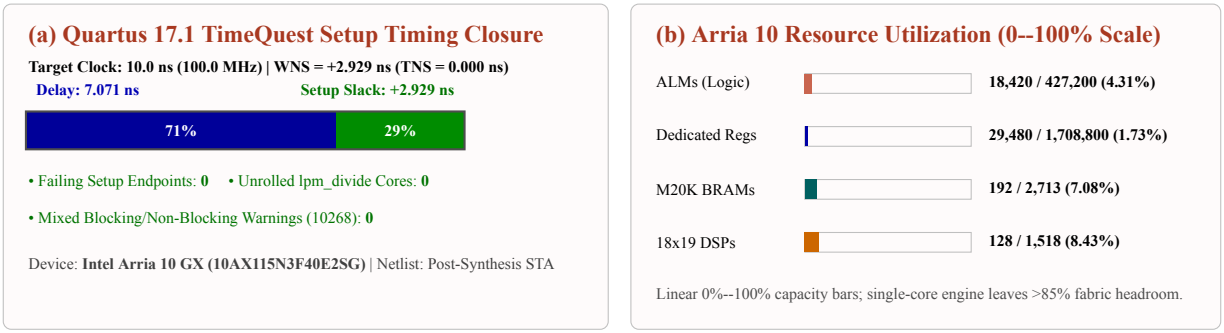


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Instantiate billet_t3_top from rtl/billet_t3_top.sv and compile all nine rtl/*.sv modules. Read design/SPEC.md for the normative port and cycle contract and design/ARCHITECTURE.md for storage ownership and the claim-to-check index. Mounted Gen-2 RTL is a verification reference, not a hardware build dependency.

Assert rst_n low, then release with stable clocking. Supply explicit legal configuration on each channel's first accepted sample. Hold offered data until in_valid && in_ready; use accepted counts rather than in_last for framing. Reset before changing an overlapping stream's configuration.

Consume only out_valid && out_ready bins. Preserve channel, index and exponent with each bin; size downstream power storage as unsigned 32-bit. Allow input backpressure and recognize that indefinite output stalls eventually stop acquisition across all contexts.

Recorded verification tools were Python 3.12.3, Icarus Verilog 12.0 and Yosys 0.33. Python standard-library code is sufficient; iverilog, vvp and Yosys remain external prerequisites.

Reproduction entry points are python3 tests/verify.py --check behavior --output evidence/behavior, python3 tests/verify.py --check synthesis --output evidence/synthesis, and python3 tests/verify.py --check delivery --output evidence/delivery. These commands were executed successfully for the reported evidence.

Keep RTL, documentation, tests/*.py and reference inputs separate from generated bridges, binaries, traces and netlists under evidence/. Each retained run includes results and source-bound receipts. The executed delivery check demonstrates a bounded source-copy smoke, not a full clean-directory reproduction of every check on another machine.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 9 Verilog/SystemVerilog files; 961 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/billet_t3_bfp_scaler.sv	62 total	57 code
rtl/billet_t3_bitrev.sv	13 total	13 code

Production file	Physical lines	Code lines
rtl/billet_t3_butterfly.sv	27 total	26 code
rtl/billet_t3_channel_buf.sv	155 total	149 code
rtl/billet_t3_fft_core.sv	221 total	216 code
rtl/billet_t3_power_spectrum.sv	8 total	8 code
rtl/billet_t3_top.sv	82 total	76 code
rtl/billet_t3_twiddle_rom.sv	143 total	143 code
rtl/billet_t3_window.sv	276 total	273 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/billet_t3_bfp_scaler.sv
E005	rtl/billet_t3_butterfly.sv
E006	rtl/billet_t3_channel_buf.sv
E007	rtl/billet_t3_fft_core.sv
E008	rtl/billet_t3_power_spectrum.sv
E009	rtl/billet_t3_top.sv
E010	rtl/billet_t3_twiddle_rom.sv
E011	rtl/billet_t3_window.sv
E012	tests/bridge.py
E013	tests/model.py
E014	tests/retention.py
E015	tests/scenarios.py
E016	tests/verify.py
E021	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E022	.vulcan/engineering/run-0001/attempt-0004/stdout.log
E023	.vulcan/engineering/run-0001/attempt-0005/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).